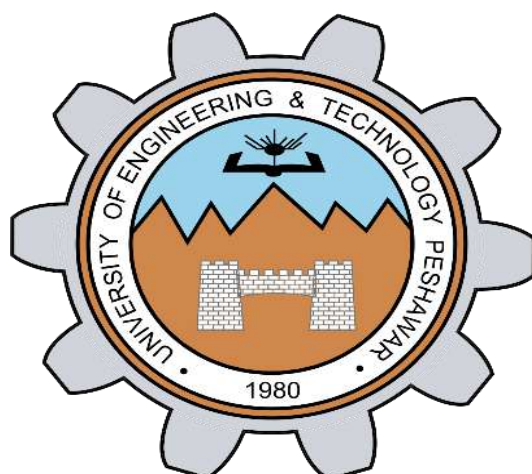


**UNIVERSITY OF ENGINEERING AND
TECHNOLOGY,PESHAWAR**



ABBOTTABAD CAMPUS

**Eco-Diplomacy: A Student-Led Circular Economy Solution
for Campus and Community Waste Management**

Submitted By:

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Course Title:

Civics And Community Engagement

Department:

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Date Of Submission:

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Phase 1

Survey of allocated sites

Allocated Areas

Your allocated survey site includes the following campus blocks:

1. Architecture Block

1. Classrooms
2. Core door areas/corridors
3. Studios

2. Electronics Department Block

1. Classrooms
2. Laboratories

3. Administration Block

4. Library

Location on map:

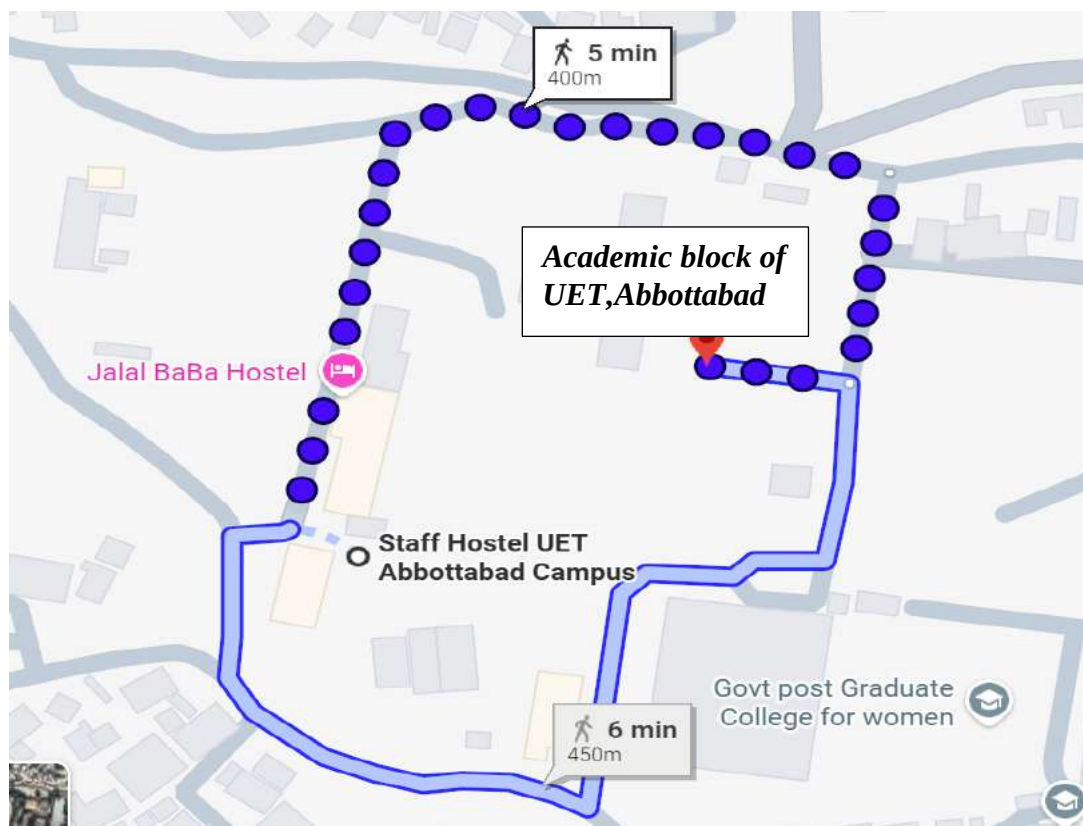


Figure 1 (red mark showing academic block of UET, Abbottabad)

1. Site Survey Documentation

Description of Surveyed Sites

Architecture Block

The Architecture Block consists of classrooms, design studios, and circulation/core areas with high student activity throughout the day. Students frequently use paper sheets, cardboard, foam boards, stationery items, and printing materials for assignments and model-making projects. Due to long studio hours, food packaging and plastic waste are also commonly generated.



Figure 2(architecture deoartment)

Electronics Department Block

The Electronics Department includes classrooms and laboratories used for practical and theoretical learning. Waste generated from this block mainly includes paper waste, plastic bottles, packaging material, damaged electronic components, wires, and laboratory-related e-waste.



Figure 3(electronic department)

Administration Block

The Administration Block contains faculty offices, administration rooms, reception areas, and official record sections. Major waste sources include printed documents, disposable cups, stationery waste, packaging material, and plastic bottles.



Figure 4(administration block)

Library

The library is a common academic space used for studying and research purposes. Waste generated here mostly consists of paper waste, snack wrappers, disposable cups, tissue paper, and plastic bottles.



Figure 5(library)



Figure 6(sitting arrangement in library)

Common Room Analysis

Short Description

The common room designated for girls was observed to be comparatively untidy during the survey. The area contained visible disposable waste and lacked proper cleanliness management.



Figure 7(girl's common room)



Figure 8(waste not thrown away in basket)

Waste Observation Table

Waste Source	Waste Type	Waste Collection Frequency
Seating Areas	Snack wrappers	1 time twice a week
Tables/Floor Areas	Paper sheets	1 time twice a week
Common Usage Areas	Disposable cups and bottles	1 time twice a week

Bin Availability

Area	Number of Bins	Type of Bin
Common Room	1 bin outside the common room	Mixed-waste dustbin



Figure 9(bin placed outside common room)

Behavior of People

- Disposable waste and wrappers were scattered in different areas.
- Paper sheets and bottles were observed on tables and floor areas.
- Cleanliness conditions were poor compared to library and administration block.
- Waste disposal practices appeared unmanaged.

Architecture Department Analysis

Short Description

The Architecture Department is one of the busiest academic areas due to studio activities and project work. Continuous student presence leads to significant generation of paper, cardboard, and food-related waste.

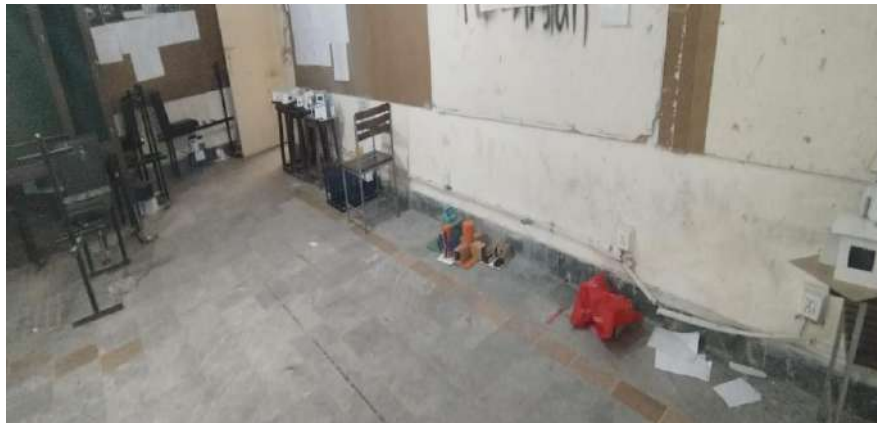


Figure 10(waste not properly disposed off)



Figure 11(corridor of architecture block)



Figure 12(careless behavior towards waste management by arhitecture students)



Figure 13(paper,card_boards thrown on ground instead of bins)



Figure 14(instead of taking their projects with them , projects are placed like waste anywhere in studios)



Figure 15(segreated waste management practice)



Figure 16 (inside condition of segregated bins)



Figure 17(papers thrown on ground,projects placed like wste,chairs missplaced)



Figure 18(corridor area)



Figure 19 (waste management practice)



Figure 20 (research lab)



Figure 21 (studio 5)



Figure 22 (store room)



Figure 23 (studio 2)



Figure 24 (room 1)

Waste Observation Table

Waste Source	Waste Type	Waste Collection Frequency
Studios	Paper sheets and cardboard	2–3 times per week
Classrooms	Plastic bottles and wrappers	2–3 times per week
Corridors/Core Areas	Mixed litter waste	2–3 times per week

Bin Availability

Area	Number of Bins	Type of Bin
Studios	No bins available	None
Classrooms	No bins available	None
Corridors	4–5	Open mixed-waste dustbins

Behavior of People

- Students usually dispose waste in corridor bins due to absence of classroom bins.
- Recycling practices are rarely observed.
- Walls of studios and corridors appear messy because students frequently use double tape to display jury work.
- Misplaced furniture creates a poor visual impression.
- Heavy use of paper and model-making materials increases waste generation.
- Corridors sometimes appear untidy during busy studio hours.

Electronics Department Analysis

Short Description

The Electronics Department contains classrooms and specialized laboratories where theoretical and practical work is carried out. The corridors are comparatively clean and organized, however classrooms show improper furniture arrangement and lack of waste bins. Dust accumulation on fans was also observed during the survey.

Laboratories Surveyed

- Analog and Communication Lab
- Final Year Project Lab
- Computer Lab 1
- Computer Lab 2
- Machine and Power Lab
- Digital Electronics Lab

Classrooms Surveyed

- Classroom 1
- Classroom 2
- Classroom 3



Figure 25 (space between architecture and electronics department)



Figure 26 (machine and power lab)



Figure 27 (computer lab 1)



Figure 28 (class room 1)



Figure 29(class room 2)



Figure 30 (class romm 3)

Waste Observation Table

Waste Source	Waste Type	Waste Collection Frequency
Laboratories	Damaged wires/components	2–3 times per week
Classrooms	Paper and plastic waste	2–3 times per week
Packaging Material	Plastic and cardboard	2–3 times per week
Corridors	Mixed waste	2–3 times per week

Bin Availability

Area	Number of Bins	Type of Bin
Laboratories	No bins available	None
Classrooms	No bins available	None
Corridors	2–3	Open mixed-waste dustbins

Behavior of People

- Students dispose waste in corridor bins due to absence of classroom bins.
- Corridors are generally clean and tidy.
- Furniture inside classrooms is not arranged in a disciplined manner.
- Dust accumulation was observed on ceiling fans.
- Awareness regarding e-waste handling is limited.
- Small electronic waste is often mixed with regular waste.

Administration Block Analysis

Short Description

The Administration Block appeared clean, tidy, and properly maintained during the survey. Offices and common areas were organized, and waste management conditions were comparatively better than academic blocks.

Waste Observation Table

Waste Source	Waste Type	Waste Collection Frequency
Offices	Paper files/documents	2–3 times per week
Staff Rooms	Disposable cups	2–3 times per week
Reception Areas	Minor plastic waste	2–3 times per week
Printing Areas	Paper scraps	2–3 times per week

Bin Availability

Area	Number of Bins	Type of Bin
Offices	Available	Small plastic bins
Corridors	Available	Mixed-waste bins
Reception	Available	Open dustbins

Behavior of People

- Staff generally maintain cleanliness.
- Furniture and office spaces are properly arranged.
- Dustbins are properly placed and used.
- Very little waste was observed during the survey.
- Waste segregation practices are absent.

Library Analysis

Short Description

The library is a well-maintained academic environment used for study and research purposes. During the survey, the area appeared clean, organized, and properly managed with minimal visible waste.

Photos to Include

- Study areas
- Properly placed dustbins
- Clean cupboards/bookshelves
- Organized seating arrangements
- Entry and circulation areas

Waste Observation Table

Waste Source	Waste Type	Waste Collection Frequency
Study Areas	Minor paper waste	2–3 times per week
Seating Areas	Minimal snack wrappers	2–3 times per week
Common Areas	Very little plastic waste	2–3 times per week

Bin Availability

Area	Number of Bins	Type of Bin
Reading Areas	Available	Mixed-waste dustbins
Entrances	Available	Open dustbins
Corridors	Available	Plastic bins

Behavior of People

- Students generally maintain cleanliness.

- Furniture is properly arranged.
 - Cupboards and bookshelves are clean and well managed.
 - Very little waste was observed during the survey.
 - Dustbins are properly placed and utilized.
-

2. Existing Waste Management Practices

Current Waste Collection Method

Waste from all blocks is collected manually by janitorial staff and transferred to central campus disposal points.

Type and Number of Bins Available

- Mostly single mixed-waste bins are available.
- Few areas have properly placed dustbins.
- Separate recycling bins are generally absent.

Waste Handling Process

Waste from classrooms, labs, offices, and corridors is collected daily and transported without segregation.

Collection Frequency

- Once or twice daily depending on building activity.
- High-traffic areas require more frequent collection.

Problems Observed

- Overflowing bins during peak hours
 - Mixing of recyclable and organic waste
 - Lack of waste segregation system
 - Littering in corridors and common areas
 - Insufficient awareness among students and staff
 - Improper handling of electronic waste in labs
-

3. Waste Identification and Quantification

Types of Waste Generated

Waste Type	Sources	Estimated Daily Quantity
Organic Waste	Food remains, tea waste	8–12 kg
Plastic Waste	Bottles, wrappers, disposable cups	10–15 kg
Paper Waste	Assignments, printouts, office files	15–20 kg
Metal Waste	Cans, small metal scraps	2–4 kg
E-Waste	Wires, damaged electronic components	1–3 kg

4. Stakeholder Identification

Stakeholder	Role	Interest	Influence
Students	Primary waste generators	Clean and comfortable environment	Medium
Faculty	Monitoring and awareness	Organized classrooms and labs	Medium
Administration	Policy and budgeting	Cost control and campus image	High
Janitorial Staff	Waste collection and cleaning	Efficient waste handling	Medium
Lab Staff	Electronic waste handling	Safety and proper disposal	Medium
Library Staff	Maintaining cleanliness	Quiet and hygienic environment	Medium

5. Environmental and Social Impact Analysis

Environmental Impacts

- Mixing recyclable waste with general waste increases landfill burden.
- Plastic waste contributes to environmental pollution.
- Improper disposal of e-waste may release hazardous substances.
- Excess paper waste results in resource loss.
- Food waste can produce odor and attract insects.

5. Environmental and Social Impact Analysis

Environmental Impacts

1. Pollution Due to Mixed Waste Disposal

The absence of a proper waste segregation system in academic blocks causes recyclable and non-recyclable waste to mix together. Paper, plastic bottles, food waste, disposable cups, wrappers, and laboratory waste are discarded in the same bins. This increases environmental pollution because recyclable materials are not recovered and instead become part of general landfill waste.

In the Architecture Department and Common Room, scattered wrappers, paper sheets, and disposable waste contribute to visual pollution and negatively affect the campus environment. Plastic waste such as bottles and wrappers is non-biodegradable and remains in the environment for long periods.

In the Electronics Department, improper disposal of wires, damaged electronic components, and packaging material may contribute to electronic waste pollution. E-waste contains materials that can become harmful to the environment if not properly managed.

2. Odor and Unsanitary Conditions

Food waste, disposable cups, and leftover snack materials observed in corridors and the common room can produce unpleasant odors if waste is not collected regularly. Waste accumulation especially in enclosed areas can create unhygienic conditions and attract insects.

Although waste is collected 2–3 times per week, areas with high student activity such as studios and common rooms may still experience temporary waste buildup. This can reduce indoor environmental quality and create discomfort for students and staff.

3. Resource Loss

Large quantities of paper waste generated in the Architecture Department, Administration Block, and Library indicate inefficient use of resources. Since paper waste is not separated for recycling, reusable materials are lost.

Similarly, plastic bottles, cardboard, and packaging materials are discarded without recovery, leading to unnecessary resource wastage. In the Electronics Department, damaged wires and electronic parts that could potentially be recycled are mixed with regular waste, reducing opportunities for sustainable waste management.

Social Impacts

1. Health Impacts

Improper waste disposal may create unhealthy conditions for students, faculty members, and janitorial staff. Dust accumulation on fans in the Electronics Department can reduce air quality and contribute to respiratory discomfort.

Waste accumulation in the common room and corridors may encourage bacterial growth, insects, and unpleasant smells, affecting hygiene standards on campus. Continuous exposure to untidy surroundings may also negatively influence students' mental comfort and productivity.

2. Cleanliness and Campus Appearance

The overall appearance of a campus strongly affects its educational environment. Misplaced furniture, messy walls with double tape marks in the Architecture Department, scattered waste in the common room, and littering in some areas create a negative visual impression.

In contrast, the Library and Administration Block demonstrate how organized furniture placement, proper dustbin usage, and cleanliness contribute to a more professional and comfortable environment.

Poor cleanliness management in certain areas can reduce the aesthetic quality of the campus and affect visitors' perception of the institution.

3. Community Complaints and User Dissatisfaction

Students and staff may become dissatisfied when waste bins are unavailable inside classrooms and studios. The need to dispose waste only in corridor bins may encourage littering and inconvenience users.

Untidy spaces such as the common room can become a source of complaints regarding hygiene and maintenance. Poorly managed waste systems may also create frustration among janitorial staff due to increased cleaning workload.

Safety Concerns

1. Risks from Electronic Waste

The Electronics Department generates small amounts of e-waste including damaged wires and electronic components. Improper disposal of such materials may expose individuals to sharp edges or hazardous substances.

Without dedicated e-waste bins or disposal procedures, these materials may become mixed with regular waste, increasing handling risks for cleaning staff.

2. Obstruction and Movement Hazards

Misplaced furniture in classrooms and studio areas may obstruct movement and reduce safe mobility inside academic spaces. Waste scattered on floors can also increase the risk of slipping or falling.

In crowded studio environments, unmanaged waste near working areas can interfere with student movement and create discomfort.

3. Hygiene and Indoor Air Quality Concerns

Dust accumulation on ceiling fans observed in the Electronics Department may affect indoor air quality and contribute to allergies or respiratory irritation.

Improperly managed waste, especially food-related waste, can attract insects and create unhygienic conditions. This may negatively affect the overall health and comfort of students and staff members using the facilities regularly.

Root Causes

Several root causes were identified during the situational audit of the Architecture Block, Electronics Department, Administration Block, Library, and Common Room.

1. Lack of Waste Segregation System

The campus does not have a proper waste segregation mechanism. Organic waste, plastic waste, paper waste, and electronic waste are disposed of together in the same bins. The absence of separate recycling bins prevents proper waste management and recycling practices.

2. Insufficient Dustbin Placement

No dustbins were available inside classrooms, studios, or laboratories in the Architecture and Electronics Departments. Students are forced to use corridor bins, which encourages littering and improper waste disposal. This issue was particularly noticeable in studios and the common room.

3. Limited Awareness Regarding Waste Management

Students and users have limited awareness regarding proper disposal practices, recycling, and cleanliness maintenance. Waste such as wrappers, disposable cups, bottles, and paper sheets was observed scattered in several areas despite the availability of corridor bins.

4. High Student Activity and Waste Generation

The Architecture Department generates large amounts of waste due to studio work, model-making activities, printing, and long working hours. Similarly, Electronics laboratories generate packaging waste and small quantities of electronic waste through practical activities.

5. Poor Maintenance Practices

Dust accumulation on ceiling fans, messy walls due to jury displays, and misplaced furniture indicate weak maintenance and monitoring systems in some departments. Regular inspection and cleanliness supervision appear insufficient in high-activity areas.

6. Lack of Dedicated E-Waste Disposal

Electronic waste generated in Electronics laboratories is mixed with general waste because no separate e-waste collection system exists. This increases environmental and safety risks.

Need for Engineering Intervention

The survey findings show a strong need for engineering-based waste management solutions to improve cleanliness, sustainability, and environmental performance on campus.

1. Installation of Segregated Waste Bins

Separate bins for:

- Paper waste
- Plastic waste
- Organic waste
- Electronic waste

should be installed in classrooms, studios, laboratories, corridors, and common rooms. Color-coded bins can improve waste segregation efficiency.

2. Smart Waste Management System

An organized waste collection and monitoring system should be developed to ensure timely waste disposal. Smart monitoring methods such as waste collection schedules, inspection checklists, or sensor-based smart bins can help prevent overflow and improve management efficiency.

3. Dedicated E-Waste Collection Mechanism

The Electronics Department requires a dedicated e-waste disposal system for damaged wires, components, and laboratory waste. Proper handling and recycling methods can reduce environmental hazards.

4. Improved Interior and Facility Management

Furniture arrangement, wall maintenance, and cleanliness monitoring should be improved, especially in Architecture studios and classrooms. Dedicated display boards can reduce wall damage caused by double tape usage during jury presentations.

5. Awareness and Educational Campaigns

Engineering intervention should also include awareness programs, posters, recycling campaigns, and student engagement activities to encourage responsible waste disposal behavior.

6. Sustainable Recycling Practices

Paper waste, cardboard, plastic bottles, and packaging materials should be collected separately and sent for recycling. This can reduce resource loss and improve campus sustainability.

7. Regular Maintenance and Hygiene Monitoring

Regular cleaning schedules, fan maintenance, and inspection systems should be implemented to maintain healthy indoor conditions and improve campus appearance.

8. Dedicated Wooden Display Mounting System

To eliminate the "dirty wall" problem caused by double-sided tape residue and paint peeling, the most professional and sustainable intervention is the installation of **fixed wooden mounting boards** across the **Architecture Block**.

Phase 2

Suggestions for waste management:

In phase 2 we wrote written proposals to chairman's of both the departments for poposing our suggestions.

1st proposal:

PROPOSAL TITLE:

Technical and Policy Proposal for Sustainable Studio Infrastructure, Material Recovery, and Visual Space Management

SUBMITTED BY:

Amina Bibi	(24abelt0988)
Iqra Noor	(24abelt1001)
Maryam Shahzad	(24abelt1003)

SUBMITTED TO:

The Chairman, Department of Architecture

Subject:

Technical & Policy Proposal for Sustainable Studio Infrastructure, Material Recovery, and Visual Space Management

- **Identified Problems (Based on Our Site Survey & Audit):**

- **Wall Damage and Tape Residue:** Severe deterioration, paint peeling, and ugly tape residue on design studio and corridor walls due to the heavy use of double-sided tape during jury presentations.



Figure 31: Tape residue and severe paint peeling on studio walls.

- **Material Waste:** Substantial loss of valuable scrap materials (foam boards, cardboards, wood off-cuts) generated during model-making activities, which are currently being thrown into general waste bins.
- **Floor Littering from Sheet Cutting:** Micro-scrap and small paper residues from manual sheet-cutting activities fall directly onto the studio floor, causing a constant mess and increasing the cleaning workload for the housekeeping staff.
- **Lack of Classroom Bins:** Complete absence of standard dustbins inside individual classrooms, leading to students stuffing litter inside or underneath their lecture desks.
- **Unsorted Corridor Waste:** Single-category bins in corridors lead to the comingling of high-grade paper waste with wet organic food waste, making recycling impossible.

- **Studio/Classroom Furniture Mismanagement:** Chairs and drafting tables are constantly left unarranged and mismanaged after classes or jury presentations, blocking walkways and creating a chaotic departmental look.



Figure 32



Figure 33: Mismanaged furniture layout and classroom littering.

- **Proposed Technical and Visual Interventions (Engineered Solutions):**

- **Fixed Wooden Display System and Live Boards:** Installation of permanent plywood, corkboard panels, or **Live Boards** across presentation walls so students can cleanly mount their drawing sheets using pins or clips instead of destructive adhesives.
- **Magnetic Paint Application:** Coating designated sections of presentation walls with high-grade **Magnetic Paint**. This allows students to attach heavy panels and blueprints cleanly using just high-strength magnets.
- **Oversized Model Scrap Bins:** Placement of specialized, large-volume bins in design studios dedicated exclusively to collecting reusable cardboard and foam board scraps.
- **Personal Desk-Attached Scrap Bins:** Attachment of small, customized waste pouches or miniature bins to every drafting workstation, capturing paper-cutting scraps immediately at the source.

- **In-Classroom Dustbins and 3-Color Corridor Hubs:** Placing standard dustbins inside every classroom, and setting up color-coded segregation hubs in corridors:
 - **Blue Bin:** Exclusively for paper and drawing materials.
 - **Yellow Bin:** For glass and plastic packaging.
 - **Green Bin:** For organic/food waste.
- **Optimized Furniture Layout Plan:** Using floor-tape or paint markings to clearly define standard workstation and chair boundaries, ensuring furniture returns to its designed position.
- **Centralized Awareness Banner (One-Stop Protocol Guide):** Designing and installing a single, large-format **Integrated Awareness Banner** at the main entry of studios/classrooms. This flex banner will crisply detail both the Waste Segregation Policy and the Studio Code of Conduct (furniture discipline and cleaning rules) in clean bullet points.
- **Social and Regulatory Action Plan (Enforcement):**
 - **"Chairs Back to Desk" Policy:** Implementing a strict policy where students must return their chairs and drafting tables to their standard marked positions before leaving the room.
 - **Strict Ban on Adhesives:** Officially prohibiting double-sided tape or liquid adhesives on painted walls, making the use of Live Boards and Magnetic walls mandatory.
 - **Student-Led Material Reuse Bank:** Forming a student eco-volunteer committee to manage the scrap bins, transforming model leftovers into a "Material Bank" for junior students.
- **Institutional and Economic Benefits for the Department:**
 - Eliminates the department's recurring annual expenditure on wall patching, plastering, and repainting.
 - Establishes a highly disciplined, visually organized, modern, and high-tech environment across studios.

2nd Proposal:

PROPOSAL TITLE:

Engineering Proposal for Hazardous E-Waste Isolation, Lab Component Management, and Library Enforcement Protocols

SUBMITTED BY:

Amina Bibi	(24abelt0988)
Iqra Noor	(24abelt1001)
Maryam Shahzad	(24abelt1003)

SUBMITTED TO:

The Chairman, Department of Electronics Engineering

Subject:

Engineering Proposal for Hazardous E-Waste Isolation, Lab Component Management, and Library Enforcement Protocols

- **Identified Problems (Based on Our Site Survey and Audit):**
 - **Hazardous Co-mingling:** Highly toxic electronic waste (damaged copper wires, burnt resistors, circuitry scrap, solder bits, capacitors) is mixed with general municipal waste inside laboratory dustbins.

- **Lab Component and Furniture Mismanagement:** Circuitry components (ICs, breadboards, jumper wires) and laboratory stools are left scattered across workstations post-practicals, causing material loss and unsafe working spaces.



Figure 34: Component mismanagement on lab workstations.



Figure 35: Mismanaged furniture layout

- **Lack of Classroom Bins:** Absence of waste bins in theoretical lecture halls, causing trash like wrappers and plastic bottles to pile up inside classrooms.
- **Unsorted Corridor Waste:** Lack of source segregation in departmental corridors, resulting in recyclable glass/plastic bottles being contaminated by organic waste.
- **Library Littering and Book Food-Damage:** Students sneaking snacks and sugary beverages inside the library, leading to pest attraction, unhygienic reading environments, and the risk of ruining rare university books.

• **Proposed Technical and Visual Interventions (Engineered Solutions):**

- **Dedicated E-Waste Collection Hubs:** Deployment of highly visible, specialized, and color-coded **E-Waste Bins** with safety warning signages strictly inside practical laboratories to achieve isolation-at-source.
- **Component Grid Storage and Workstation Setup:** Introducing labeled modular grid drawers/boxes for easy sorting of electronic parts, alongside fixed layout boundaries for lab stools and testing equipment.

- **In-Classroom Dustbins and 3-Color Corridor Hubs:** Introducing standard bins inside lecture rooms and installing color-coded sorting hubs in corridors:
 - **Blue Bin:** For paper waste and documentation scrap.
 - **Yellow Bin:** For glass and plastic bottles.
 - **Green Bin:** For organic/food waste.
- **Library Hygiene Stations:** Placing high-capacity disposal hubs at the entrance of the library, and mapping out a strict zero-litter zone inside.
- **Centralized Awareness Banner (Lab and Safety Protocol Guide):** Mounting a prominent **Integrated Awareness Banner** outside major practical labs and lecture halls. This single banner will clearly outline E-Waste Disposal Steps and Lab Discipline Rules (returning components to correct grid drawers and tucking chairs neatly under workstations).
- **Social and Regulatory Action Plan (Enforcement):**
 - **Laboratory Code of Conduct:** Making it mandatory for students to clear their testing tables, store components in their respective grids, and arrange chairs before lab sessions are officially dismissed by instructors.
 - **Strict Library "No-Food" Policy:** Setting up clear warning signages at the library entrance and empowering the entry desk staff to physically stop students from bringing food items or beverages inside.
 - **Orientation Briefings:** Conducting short 10-minute mandatory orientation sessions at the start of the semester to educate students on electronic hazards and layout discipline.
- **Institutional and Economic Benefits for the Department:**
 - Mitigates any campus liability regarding toxic heavy-metal contamination or open dumping hazards.
 - Reduces laboratory equipment/component losses through systematic grid storage, while preserving the clean and peaceful environment of the library.

Phase 3:

Eco-Diplomacy Meetings' Report

1. Introduction:

- In Phase 3 of this project, our group focused on "Eco-Diplomacy." This means we took the garbage and space problems we found in our Phase 1 Audit Report and went to discuss them with the university authorities.
- Our main goal was to present our ideas to the management, get their feedback, and find a realistic way to fix the cleanliness and discipline issues on campus. During this process, we faced some real-world challenges, funding problems, and administrative limits. This report shows who we met, what they said, and how we changed our plan to make it work without spending money.

2. Objectives of Phase 3:

- To present our customized waste and space management plans to the university authorities.
- To understand the administrative problems and budget limits of our campus.
- To find zero-budget, low-cost solutions since the university cannot provide funds.
- To involve helpful departments, like the Facilitation Centre, to support our cleanup plan.
- To create a step-by-step implementation plan that students can manage by themselves.

3. Stakeholders Engaged:

We mapped out and met the following key people in the university to discuss our proposals:

Table 1: Stakeholders and Their Administrative Roles

Name	Department	Role in Our Project
Chairman, Department of Architecture	Academic Department	Reviewed the proposal but directed us to General Administration for cleaning issues.
Chairman, Department of Electronics	Academic Department	Reviewed the lab e-waste problem but directed us to General Administration.
Mr. Faisal Nawaz	Central Administration	In charge of campus cleanliness. Shared his past efforts and explained the lack of funds.
Facilitation Centre Staff	Campus Services	Welcomed our ideas and explicitly agreed to cooperate with our team for implementation.

4. Real Account of the Meetings (What Happened):

4.1 Meeting with the Chairmen (Architecture & Electronics):

First, we went to the Chairman of Architecture and the Chairman of Electronics Engineering with our written proposals. We showed them the problems like tape marks on walls and loose electronic waste. Both Chairmen gave us a direct answer: "This is not our job. For cleanliness and maintenance, go to the Admin Officer, Mr. Faisal Nawaz."

4.2 Meeting with the Administration (Mr. Faisal Nawaz):

Next, we met Mr. Faisal Nawaz. He was very honest with us and explained that the administration has worked very hard on campus cleanliness in the past, but the garbage problem always returns.

- **The Letters:** He showed us official letters that his office had written to the Assistant Commissioner (AC) and the local municipal cleaning teams asking for help. Unfortunately, those external teams never responded or helped. The campus administration feels helpless ("hands up") due to this lack of support.
- **The Funding Problem:** When we suggested buying 3-color corridor bins and standard dustbins for classrooms, he indirectly refused due to a total lack of funds. He also told us that according to university policy, we cannot collect money from students to buy these bins.

4.3 Meeting with the Facilitation Centre:

After the budget disappointment, we met with the team at the Facilitation Centre. This meeting was highly positive. The staff appreciated our initiative and clearly told us that they will fully cooperate with us to help implement our solutions on campus. Their support gives our project a fresh pathway to succeed.

5. Re-Engineered Zero-Budget Solutions (Our Final Plan):

Since the administration has no money and we cannot take funds from students, we changed our design. We dropped all expensive ideas (like commercial bins, wooden boards, or costly magnetic paint). Instead, we created Zero-Cost, Student-Led Solutions based on the 5R framework:

5.1 Refuse (Zero Cost):

- **Official Ban on Tapes:** The administration can issue a free notification to ban double-sided tape and liquid glue on painted studio walls. This costs zero rupees but will save the walls from damage.
- **Library No-Food Rules:** Setting up basic warning signs at the library entrance. The entry desk staff can stop students from bringing food and sweet drinks inside.

5.2 Reduce & Reuse via Upcycling (Zero Cost):

- **DIY Carton Bins:** Since we cannot buy bins, our student team will collect large, empty cardboard boxes and plastic containers from campus juice stalls and shops. Student volunteers will paint and label them to make homemade dustbins for classrooms and corridors.
- **Cardboard Material Banks:** We will place large upcycled boxes inside Architecture studios labeled "Material Reuse Bank." Students can drop their leftover modeling pieces (foam boards, chart papers) here so junior students can reuse them for free.

5.3 Recycle & Lab Rules (Zero Cost):

- **Self-Service Component Grid:** In the Electronics labs, a strict rule will require students to put jumper wires, ICs, and breadboards back into the existing lab drawers after practicals. This stops expensive engineering tools from being thrown in the trash.
- **3-Color Upcycled Hubs:** We will arrange our homemade carton bins in corridors in sets of three:
 - **Blue Box:** For clean waste paper and sheets.
 - **Yellow Box:** For plastic bottles and snack wrappers.
 - **Green Box:** For leftover food and tea waste.

5.4 Spatial Discipline (Zero Cost):

- **Floor Layout Markings:** We will use existing markers or chalk to draw simple placement lines on studio floors for drafting tables and chairs.
- **Digital Awareness Campaign:** Instead of printing expensive flex banners, we will design free digital posters and share them across student WhatsApp groups and campus screens to explain the cleanliness rules.

6. Technical Framework & Execution Strategy:

Our new plan completely bypasses the funding issue and the silence of external municipal authorities:

- **No Money Needed:** By using garbage items (scrap cartons and plastic bottles) and student volunteer help, we do not need a single rupee from the university budget.
- **Facilitation Centre Support:** We will use the promised cooperation of the Facilitation Centre to help coordinate our work and communicate smoothly with the cleaning staff.
- **Future Student Involvement:** Since we have not discussed this with the general student body yet, our next immediate step is to pitch this as a peer-to-peer responsibility, showing them how the Material Bank saves them money.

7. Proposed Execution Timeline:

- **Phase 1 (Immediate):** Administration issues the notice to ban wall tapes. We start the free digital awareness drive on student groups.
- **Phase 2 (Mid-Week):** Collect waste cartons from campus shops. Student volunteers paint and prepare them into labeled classroom bins.
- **Phase 3 (End of Week):** Place the upcycled bins in lecture rooms and corridors. Set up the Cardboard Material Banks in design studios.
- **Phase 4 (Ongoing):** Work closely with the Facilitation Centre and cleaning staff to ensure the upcycled bins are emptied daily.

8. Roles and Responsibilities:

Table 2: Stakeholders and Zero-Cost Responsibilities

Stakeholder Group	Responsibility (No Budget Required)
Campus Admin (Mr. Faisal Nawaz)	Issuing the official ban on wall tapes and giving permission for upcycled bins.
Facilitation Centre Team	Coordinating between our group, students, and janitorial staff to keep the system moving.
Our Project Team	Organizing the box collection drive, making the DIY bins, and running digital campaigns.
Lab Attendants	Asking students to clean their lab desks and return components before leaving.
Housekeeping Staff	Emptying our homemade upcycled corridor boxes during their regular daily cleaning rounds.
Student Body (Proposed)	Keeping their own tables clean, using the material banks, and putting chairs back on floor lines.

9. Expected Outcomes:

- **Zero Cost System:** A fully working garbage sorting system made completely from upcycled materials without spending any money.
- **Immediate Wall Protection:** Stopping paint peeling in Architecture studios by implementing the tape ban.
- **Help for Cleaning Staff:** Classroom litter will decrease because every room will finally have an upcycled bin, reducing the workload of janitors.
- **Resource Saving:** Students will save money by sharing modeling scraps through the Cardboard Material Bank.

10. Conclusion:

- Our Phase 3 meetings taught us a lot about real-world management problems. When the Chairmen shifted the responsibility, the AC office did not reply, and the administration had zero budget, it looked like a dead end.
- However, by getting the cooperation of the Facilitation Centre and turning trash (old cartons) into useful dustbins, we found a practical way forward. This plan proves that with student unity and smart upcycling, UET Peshawar's Abbottabad campus can become clean, organized, and disciplined without depending on external funds.

Phase 4:

1. IMPLEMENTATION OF THE APPROVED WASTE MANAGEMENT SOLUTION

Departmental Leadership Engagement

As part of the implementation phase, the project team prepared and submitted formal written applications containing practical waste management recommendations to the departmental leadership. Meetings were held with Sir Adam (Chairman, Electronics Engineering Department) and Sir Shahid (Chairman, Architecture Department). Both chairmen appreciated the initiative and acknowledged the importance of improving waste management practices within the university. However, they informed the team that the departments were facing financial limitations and therefore could not provide funding or material resources for implementation.

Despite these limitations, both departments expressed moral support for the project and encouraged student-led efforts to improve environmental practices.

Signed proposals :

The proposals submitted in phase 2 got signed in phase 4 and we received healthy remarks.

Proposal signed by sir Adam khan(chairman of electronics department):

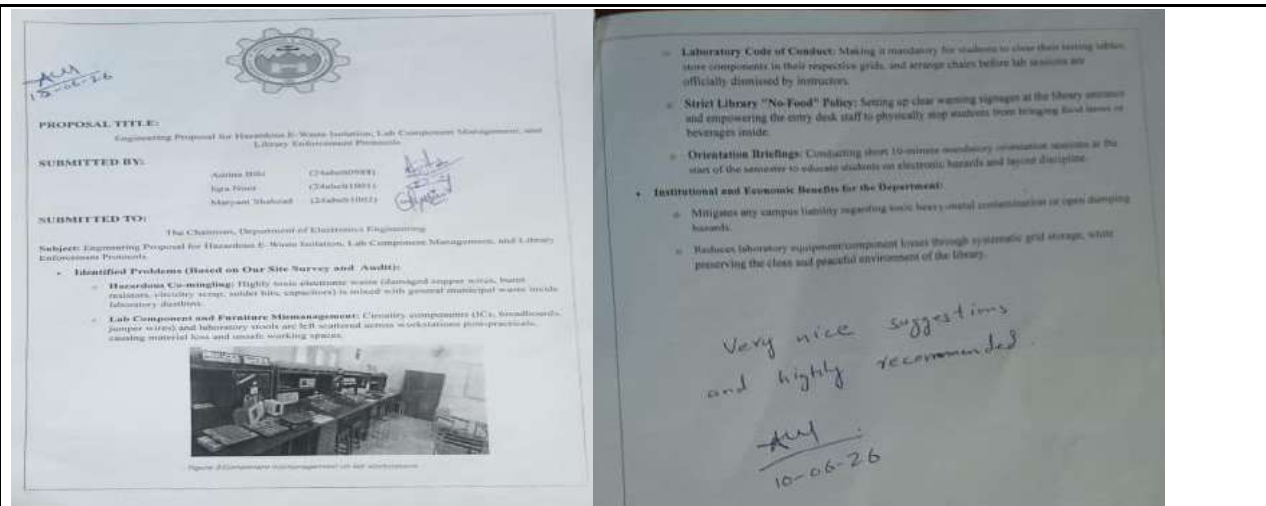


Figure 36(proposal signed by chairman of electronic engineering with satisfactory remarks)

Proposal signed by sir Shahid (chairman of Architecture department):

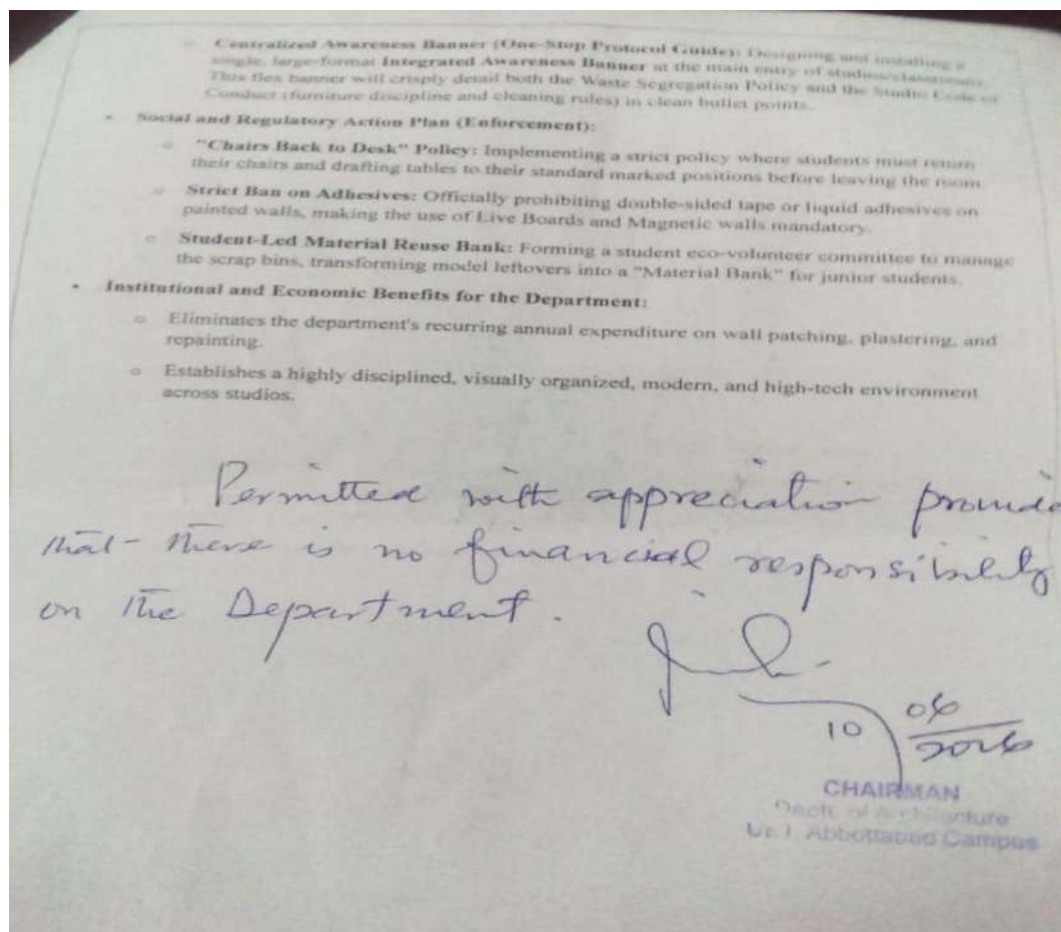


Figure 37(proposal signed by chairman of architecture engineering with satisfactory remarks)

Administrative and Municipal Assessment

To understand the broader waste management situation, the project team held detailed discussions with Sir Faisal, Head of Administration.

During these discussions, Sir Faisal revealed that the university's waste management challenges extend beyond student behavior and internal cleaning operations. He explained that one of the major issues affecting campus cleanliness is the failure of municipal waste collection services to regularly lift and transport accumulated waste from the university.

To support his statements, Sir Faisal shared copies of official correspondence addressed to regional waste management authorities regarding these collection failures. He also provided access to a formal letter submitted to the Assistant Commissioner (AC), requesting intervention and assistance in resolving the waste disposal crisis.

The review of these documents demonstrated that the waste problem is not solely an institutional issue but also a municipal governance challenge affecting the wider community.

Administrative Challenges

The project team attempted to engage Sir Akhtar Muneer, the campus coordinator, to obtain centralized institutional support and discuss potential campus-wide implementation.

However, the request was declined. He informed the team that due to a previous incident involving inappropriate student behavior, he was no longer permitting students to enter his office for discussions.

As a result, centralized administrative support could not be secured.

Adaptive Implementation Strategy

In response to these challenges, the project adopted a decentralized implementation approach.

Instead of relying on central administration, the team worked directly with departmental stakeholders. Through the cooperation of Ma'am Iqra from the Architecture Department, access was secured for Studio 1, while Electronics Lab was selected as the second pilot location.

This approach allowed implementation activities to continue despite administrative barriers.

2. INSTALLATION OF PHYSICAL WASTE MANAGEMENT SYSTEMS

Resource-Constrained Implementation

The university did not provide any financial assistance, equipment, or budget for the project.

Consequently, the implementation relied entirely on student initiative, existing resources, and low-cost solutions. The project team utilized available containers and existing infrastructure to establish basic waste segregation points without requiring additional institutional expenditure.

Site Preparation

Before implementing the pilot waste segregation system in Architecture Studio 1, the project team directly approached Abid Bhai, a member of the university's cleanliness staff, and explained the objectives of the Eco-Diplomacy project.

The team requested his assistance in cleaning and preparing Studio 1 so that waste management activities could begin in a clean and organized environment. After understanding the purpose of the initiative, Abid Bhai willingly agreed to support the project.

He subsequently carried out the cleaning of Studio 1, removing accumulated waste and helping create a suitable environment for the implementation of the waste segregation system.

This cooperation demonstrated that although the project received no financial assistance from the university, meaningful support could still be obtained through collaboration with operational staff who recognized the value of improving campus cleanliness.

The involvement of Abid Bhai became an important example of grassroots stakeholder engagement, showing how sustainability initiatives can succeed through cooperation between students and support staff even when institutional resources are limited.



Figure 38(talking to our stalkholder abid bhai)

Deployment of Segregation Points

Waste segregation points were established at:

- Architecture Studio 1
- Electronics Lab

These locations were selected because they generate significant amounts of recyclable material, including cardboard, chart paper, packaging materials, and laboratory waste.

Separate collection areas were designated to encourage waste sorting and reduce contamination.

Awareness-First Approach Instead of Printed Signage

No posters, banners, or printed awareness materials were installed. This decision was intentional and based on both environmental and practical considerations.

First, producing additional printed materials would generate unnecessary waste, which would contradict the project's sustainability objectives.

Second, the team observed that students often ignore static posters after a short period, reducing their effectiveness as long-term behavioral interventions. Instead, awareness was promoted through direct communication, classroom presentations, peer-to-peer engagement, and continuous interaction with students. This approach encouraged active participation while maintaining the project's zero-budget mode

3. AWARENESS SESSIONS AND STUDENT TRAINING

Faculty Collaboration

Faculty members from both departments showed strong cooperation throughout the implementation phase.

Several instructors voluntarily allocated time during their classes to allow project team members to introduce the initiative and explain its objectives.

This support significantly improved student outreach and engagement.

Classroom Awareness Sessions

Awareness sessions were conducted for first-year students in both departments.

The sessions focused on:

- Environmental impacts of improper waste disposal.
- Principles of waste segregation.
- Circular economy concepts.
- Student responsibilities in maintaining clean learning environments.
- The importance of reducing contamination in recyclable waste streams.

The team used interactive discussions rather than formal lectures to encourage participation and understanding.

Training in Architecture Studio 1

Architecture students were guided on:

- Separating clean cardboard and modeling materials from food waste.
- Preserving reusable design materials.
- Maintaining the cleanliness of recyclable waste streams.

Because Architecture students generate large quantities of cardboard, chart paper, and model-making materials, proper segregation significantly improved recycling potential.

Training in Electronics Lab

Electronics students were instructed to:

- Separate packaging materials from mixed waste.
- Properly dispose of laboratory-related waste.
- Prevent recyclable materials from becoming contaminated.



Figure 39(training electronics' students)

Ground Staff Cooperation

Although financial support was unavailable, the cooperation of the cleaning staff proved valuable.

Abid Bhai actively supported the initiative by maintaining the designated areas and helping ensure that segregated materials remained separated during routine cleaning activities.

4. FIELD TESTING AND MONITORING

Monitoring Activities

To evaluate implementation effectiveness, the project team conducted regular inspections of both pilot locations.

Monitoring visits focused on:

- Proper placement of segregation points.
- Student awareness of waste sorting procedures.
- Compliance with segregation guidelines.
- Levels of contamination.
- Waste accumulation patterns.
- Effects of municipal collection failures on campus operations.

Monitoring Checklist

Observation Item	Status	Remarks
Segregation point correctly placed	Yes	Accessible to students in both locations
Students aware of segregation instructions	Yes	Awareness created through direct communication
Verbal guidance provided by team members	Yes	Conducted regularly
Waste properly segregated	Mostly Yes	Minor contamination observed during busy periods
Overflow observed	Occasionally	Increased waste generation after practical sessions
Municipal collection bottleneck observed	Yes	Campus waste accumulation remained visible
Student participation observed	Yes	Positive response from first-year students

Municipal Collection Monitoring

Throughout the monitoring period, observations supported concerns previously raised by Sir Faisal.

Although internal waste segregation improved, delays in municipal waste collection continued to create congestion at secondary campus disposal points.

This finding demonstrated that effective campus sustainability efforts require both internal management improvements and reliable municipal support.

5. OPERATIONAL DATA COLLECTION AND OBSERVATIONS

Quantitative Waste Tracking

Waste quantities were periodically measured and recorded to understand the composition of waste generated within the pilot areas.

Sample Waste Audit Data

Date	Location	Organic Waste (kg)	Recyclable/Studio Scrap (kg)	Residual Waste (kg)	Observations
Day 1	Electronics Lab	3.8	1.5	2.2	Strong student participation due to faculty support
Day 1	Architecture Studio 1	1.5	9.8	3.0	High quantity of clean cardboard and chart paper
Day 3	Electronics Lab AB	4.2	1.2	2.5	Internal segregation functioning effectively
Day 3	Architecture Studio 1	2.0	12.4	4.1	Large volume of recyclable studio materials generated

Observation of studio 1:

Situation of studio 1 before our work:



Figure 40(Papers thrown on ground instead of using dust bin)

Situation of studio 1 After our work:

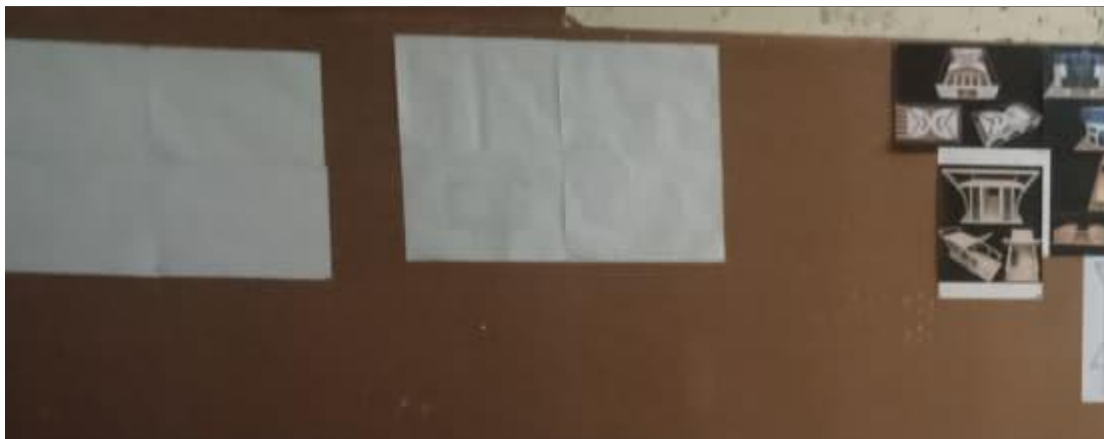


Figure 41(everything is placed in a mannered way to reduce waste)



Figure 42(Papers are disposed off in dust bin)



Figure 43(Installation of proper segregated bins)

Observation of Electronics lab:

Situation of Electronics Lab before our work:



Figure 44(Wires are not placed back to their place after use)

Situation of Electronics Lab after our work:



Figure 45(Wires placed back to the tray for avoiding waste)

Key Findings

Internal Strengths

- Strong support from departmental leadership.
- Excellent cooperation from faculty members.
- Positive engagement from first-year students.
- Valuable assistance from cleanliness staff.
- Successful implementation despite the absence of funding.

Major Challenges

- No financial support from the university.
- Limited access to centralized administration.
- Municipal waste collection failures.
- Dependence on external authorities for final waste removal.

MACRO-LEVEL POLICY AND GOVERNANCE REVIEW

One of the most important discoveries during Phase 4 was that waste management challenges on campus are connected to broader governance and municipal service delivery issues.

The official correspondence shared by Sir Faisal demonstrated that university administration has repeatedly raised concerns regarding inadequate waste collection services.

The formal letter addressed to the Assistant Commissioner further highlighted the need for government intervention to resolve waste transportation and disposal challenges.

As a result, the project evolved beyond a simple campus waste segregation initiative. It became an exercise in environmental governance, civic engagement, and eco-diplomacy by examining how educational institutions interact with municipal authorities in addressing sustainability challenges.

EXPECTED OUTPUTS ACHIEVED

- ✓ Submission of formal waste management recommendations to departmental leadership.
- ✓ Successful establishment of pilot waste segregation systems in Architecture Studio 1 and Electronics Lab AB.
- ✓ Completion of awareness sessions through direct communication and faculty-supported classroom engagement.
- ✓ Collection of photographic and video evidence documenting implementation activities.
- ✓ Documentation of municipal waste collection failures affecting campus sustainability efforts.
- ✓ Review and analysis of official correspondence between university administration and regional waste management authorities.
- ✓ Identification of policy-level barriers impacting environmental management within the university.

CONCLUSION

Phase 4 successfully demonstrated that meaningful environmental action can be achieved even in the absence of institutional funding. Through direct communication, stakeholder engagement,

faculty cooperation, and student participation, the project established functioning waste segregation practices within the Academic Block.

The implementation also revealed that campus waste management challenges are closely linked to larger municipal governance issues. While internal segregation systems proved effective, long-term sustainability will require stronger collaboration between the university and external waste management authorities.

The findings and data collected during this phase provide a strong foundation for Phase 5, which will focus on impact evaluation, stakeholder recommendations, policy advocacy, and the development of a scalable circular economy framework for the university and surrounding community.

PHASE 5

Activity: Assessing the Effectiveness of the Waste Management Solution

Following the implementation of the waste management solution in the selected pilot locations, the project team evaluated its effectiveness through direct observations, waste audits, stakeholder feedback, and regular monitoring activities. The evaluation focused on determining whether the implemented interventions improved waste segregation practices, increased environmental awareness, and promoted sustainable waste management behaviour among students and staff.

1. Measuring the Effectiveness of the Implemented Waste Management Solution

The evaluation focused on the effectiveness of the pilot waste segregation system, student participation in waste segregation activities, awareness sessions conducted within the Architecture Department and Electronics department, stakeholder cooperation, and improvements in waste disposal practices.

The findings indicated that the intervention was successful despite the absence of institutional funding. Students actively participated in waste segregation after receiving awareness sessions, and recyclable materials such as cardboard, chart paper, packaging materials, and laboratory waste were more frequently separated from mixed waste. Faculty members supported the initiative by allowing awareness sessions during class time, while operational staff cooperated in maintaining the designated waste segregation areas.

Overall, the project demonstrated that low-cost, student-led interventions supported by stakeholder collaboration can significantly improve waste management practices within academic environments.

2. Analysis of Monitoring Data

Table 1: Analysis of Monitoring Data

Indicator	Before Implementation	After Implementation
Mixed waste in pilot locations	High	Moderate
Properly segregated waste	Limited	Frequently Observed
Student participation in waste segregation	Low	High
Student awareness of waste segregation	Moderate	High
Faculty support for awareness activities	Limited	Active
Cleaning staff cooperation	Routine	Active Support

The monitoring results indicate noticeable improvements in waste segregation behaviour, environmental awareness, stakeholder participation, and overall cleanliness within the pilot locations. Although occasional contamination of segregated waste was observed during busy laboratory sessions, the overall effectiveness of the waste management system improved significantly compared to the pre-implementation conditions.

3. Comparison of Before and After Conditions

Before implementation, waste generated in the Architecture Department and Electronics department was generally disposed of as mixed waste without any systematic segregation. Recyclable materials such as cardboard, chart paper, packaging materials, and laboratory waste were commonly mixed with general waste. Students had limited awareness regarding waste segregation, and dedicated waste segregation points were not available.

After implementation, waste segregation points were established in the selected pilot locations. Students actively participated in separating recyclable materials from general waste, and awareness sessions improved understanding of responsible waste disposal practices. Architecture students preserved reusable model-making materials, while Electronics students demonstrated improved handling of laboratory-related waste by returning reusable components to their designated storage locations. These improvements resulted in cleaner working environments and enhanced waste management practices.

4. Evaluation of Environmental, Social, and Operational Impacts

Environmental Impacts

The project produced several positive environmental outcomes. Waste segregation improved the recovery potential of recyclable materials and reduced contamination of recyclable waste streams. Better handling of laboratory waste minimized the likelihood of recyclable materials being mixed with general waste. The implementation also contributed to cleaner academic spaces and promoted environmentally responsible behaviour among students.

Social Impacts

The project strengthened environmental awareness among students and encouraged active participation in sustainability initiatives. Collaboration between students, faculty members, departmental leadership, and operational staff fostered a stronger sense of shared responsibility for maintaining a clean and sustainable campus environment. Increased student engagement also contributed to improved environmental stewardship within the participating departments.

Operational Impacts

Operationally, the project improved the organization of waste collection activities within the pilot locations. Waste segregation points made waste disposal more systematic and supported the cleaning staff during routine waste collection. Faculty cooperation enhanced awareness efforts, while regular monitoring enabled the project team to identify challenges and recommend improvements. The implementation demonstrated that meaningful improvements can be achieved through stakeholder collaboration even when financial resources are limited.

5. Reflection on Sustainability Policies and Improvement Needs

The evaluation demonstrated that effective waste management depends on continuous stakeholder engagement, institutional commitment, and regular monitoring rather than one-time interventions. Although significant improvements were achieved through the pilot implementation, several areas require further development.

The university currently lacks a comprehensive campus-wide waste segregation policy and dedicated procedures for managing recyclable and laboratory waste. Financial limitations restricted the expansion of waste segregation infrastructure beyond the pilot locations, while delays in municipal waste collection continued to affect overall waste management performance.

To ensure long-term sustainability, waste segregation practices should be expanded across all university departments, environmental awareness activities should become part of student orientation programmes, regular waste audits should be conducted, and stronger coordination should be established between the university administration and municipal waste management authorities.

6. Recommendations for Future Actions

1. Expand waste segregation facilities to all academic departments, laboratories, libraries, and common areas.
 2. Conduct regular environmental awareness campaigns and workshops throughout the academic year.
 3. Establish a permanent Green Committee comprising students, faculty members, and administrative representatives to oversee campus sustainability initiatives.
 4. Introduce dedicated collection systems for laboratory-related recyclable waste and electronic waste.
 5. Conduct periodic waste audits and monitoring activities to evaluate progress and identify areas requiring improvement.
 6. Strengthen collaboration between the university administration and municipal waste management authorities to improve external waste collection services.
 7. Develop a digital monitoring and reporting system for tracking campus waste management performance.
 8. Continue encouraging student-led environmental initiatives that promote sustainable waste management practices across the university.
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Conclusion

The evaluation indicates that the Eco-Diplomacy Project successfully improved waste segregation practices, increased environmental awareness, strengthened stakeholder collaboration, and promoted sustainable waste management within the selected pilot locations. Despite financial constraints and external municipal challenges, the project demonstrated that student-led initiatives supported by faculty and operational staff can generate meaningful environmental improvements. Continued monitoring, institutional support, and stakeholder participation will be essential for sustaining these achievements and expanding the initiative across the university.

PHASE 6

Activity: Reflection on Project Experience

At the conclusion of the Eco-Diplomacy Project, the project team conducted a structured reflection exercise to evaluate the overall project experience and identify the key lessons learned throughout the planning, implementation, monitoring, and evaluation phases. Team members reflected on their individual roles, contributions, challenges encountered, and the impact of the project on their understanding of sustainability, environmental responsibility, and engineering practice.

The project provided valuable practical experience in identifying environmental problems, conducting situational audits, engaging stakeholders, preparing technical proposals, implementing sustainable waste management solutions, and evaluating their effectiveness within a real university environment. Working collaboratively with departmental leadership, faculty members, operational staff, and fellow students significantly strengthened communication, teamwork, leadership, and problem-solving skills.

1.Evaluation of Personal and Professional Learning Outcomes

The project contributed substantially to both personal and professional development. Team members developed a deeper understanding of sustainable waste management systems, environmental stewardship, and the importance of integrating sustainability into engineering practice.

Students also enhanced their competencies in environmental auditing, stakeholder consultation, proposal writing, project planning, implementation, data collection, monitoring, analysis, and technical report writing.

From a professional perspective, the project strengthened engineering decision-making, teamwork, leadership, communication, and evidence-based problem-solving skills. It also demonstrated that engineering solutions must balance technical, environmental, economic, and social considerations to achieve sustainable outcomes.

2.Key Learning Outcomes

- Improved understanding of sustainable waste management systems and environmental sustainability principles.
 - Enhanced knowledge of environmental auditing and waste assessment techniques.
 - Strengthened communication and stakeholder engagement skills.
 - Improved teamwork, leadership, and collaborative decision-making abilities.
 - Increased ability to analyse environmental problems using engineering approaches.
 - Enhanced problem-solving and critical-thinking skills.
 - Greater appreciation of evidence-based engineering decision-making and professional ethics.
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3.Successes, Challenges, and Unexpected Issues

3.1 Successes

The project successfully established pilot waste segregation systems in Architecture Studio 1 and the Electronics Laboratory despite the absence of institutional funding. Formal proposals

received positive remarks from both departmental chairmen, awareness sessions reached first-year students through faculty support, and operational staff actively cooperated during implementation.

The project also successfully documented municipal waste management challenges and demonstrated that low-cost, student-led sustainability initiatives can positively influence environmental practices within academic institutions.

3.2 Challenges

Several challenges affected project implementation. Financial constraints prevented the installation of additional waste segregation infrastructure throughout the campus. Limited access to centralized administration required the project team to adopt a decentralized implementation strategy. Furthermore, delays in municipal waste collection continued to affect overall campus waste management despite improvements achieved within the pilot locations.

3.3 Unexpected Challenges

Occasional contamination of segregated waste occurred during periods of increased student activity, requiring repeated awareness efforts and continuous monitoring. Maintaining consistent student participation throughout the implementation period also proved challenging. These experiences highlighted the importance of ongoing environmental education and stakeholder engagement for sustaining behavioural change.

4. Integration of Engineering Practice with Sustainability and Societal Responsibility

The project demonstrated that engineering extends beyond technical problem-solving and plays an important role in addressing environmental and social challenges. Through environmental auditing, stakeholder engagement, proposal development, implementation, monitoring, and evaluation, students experienced how engineering knowledge can be applied to promote sustainable development and improve community well-being.

The project reinforced the professional responsibility of engineers to design practical, cost-effective, and environmentally sustainable solutions while considering technical feasibility, economic limitations, social acceptance, and long-term environmental impacts.

5. Future Improvements and Scalability Plans

Based on the experience gained during the project, the team recommends the following future improvements:

1. Expand waste segregation systems to all university departments, laboratories, hostels, libraries, and common areas.

2. Establish permanent student-led Green Committees to coordinate environmental activities.
 3. Develop a digital waste monitoring and reporting system to improve performance evaluation.
 4. Introduce dedicated collection systems for laboratory waste and electronic waste throughout the university.
 5. Conduct regular environmental awareness programmes and sustainability workshops for students and staff.
 6. Strengthen collaboration between university administration and municipal waste management authorities.
 7. Integrate sustainability education into student orientation programmes.
 8. Replicate the Eco-Diplomacy waste management model in other departments and university campuses.
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Conclusion

The reflection process demonstrated that the Eco-Diplomacy Project successfully integrated engineering practice with sustainability, stakeholder engagement, and societal responsibility. Despite financial limitations and administrative challenges, the project achieved measurable improvements in waste segregation, environmental awareness, and campus cleanliness through collaborative, low-cost interventions. The experience strengthened the technical, professional, and leadership competencies of the project team while providing a practical and scalable framework for promoting sustainable waste management across the university.